2026-10-01 | VS&A; DAILY | 10-MINUTE LESSON

Legacy sine sweeps versus random fatigue

Learning objective. Convert a legacy swept-sine and resonance-dwell test into a defensible fatigue comparison with a random environment - at the hardware response, not at the input label.

A 60-year-old test sheet may specify double-amplitude displacement, an acceleration plateau, a logarithmic sweep rate, “up and back,” and several resonance dwells. A current requirement may be a PSD and duration. There is no universal conversion between those descriptions. They excite a structure differently and distribute cycles differently.

Yet this is not a new problem. NASA CR-242, published in June 1965, treated sine sweep, sine dwell and random loading as separate damage contributors. It converted each into response cycles, evaluated damage against fatigue behavior, and accumulated the contributions [1]. The lasting lesson is methodological: **make the comparison after structural filtering.**

Legacy sine schedule → response amplitude versus frequency → cycles at each amplitude
Random PSD → response PSD or time history → statistical or counted stress cycles

The worked example below is original and illustrative. It is not a reconstruction of a named vehicle, a certification rule or measured telemetry. Its purpose is to show the bookkeeping and the assumptions that can change the answer.

A good comparison can support qualification-by-analysis, test tailoring or a heritage assessment. It cannot prove that a new test covers unrelated objectives such as workmanship, functional operation, peak load, joint slip or control stability.

The comparison should answer a bounded decision: whether a particular detail and failure mode receives at least the intended fatigue exposure. It should not begin by asking for a single “equivalent g level.” That shortcut hides modal placement, damping and duration - exactly the quantities that control the response. Where original test recordings survive, they are more valuable than the specification envelope because they preserve achieved levels, notches, dwell behavior and interruptions. Where only paper records remain, bracket every ambiguous convention and carry the alternatives through the calculation.

01 / DECODE THE OLD TEST

Double amplitude is the first trap

Legacy sinusoidal specifications often used constant displacement at low frequency and constant acceleration above a crossover. If DA is peak-to-peak displacement in inches, the zero-to-peak acceleration is

$$a_{pk}/g = (2\pi f)^2(DA/2)/g = 0.0511\ f^2\ DA$$

Calling the DA value an amplitude, or treating acceleration as double amplitude, creates an immediate factor-of-two error. With a fatigue slope m = 5, a factor-of-two stress error becomes a factor of 32 in nominal damage.

The illustrative schedule uses 0.060 in DA from 5 to 20 Hz, then 1.2 g peak to 200 Hz, one logarithmic sweep up and one down at 2 octaves/min, followed by a 60 s dwell at a measured 100 Hz resonance. The hardware is represented by one linear mode with 2% damping and an effective stress/acceleration gain K = 2 MPa/g.

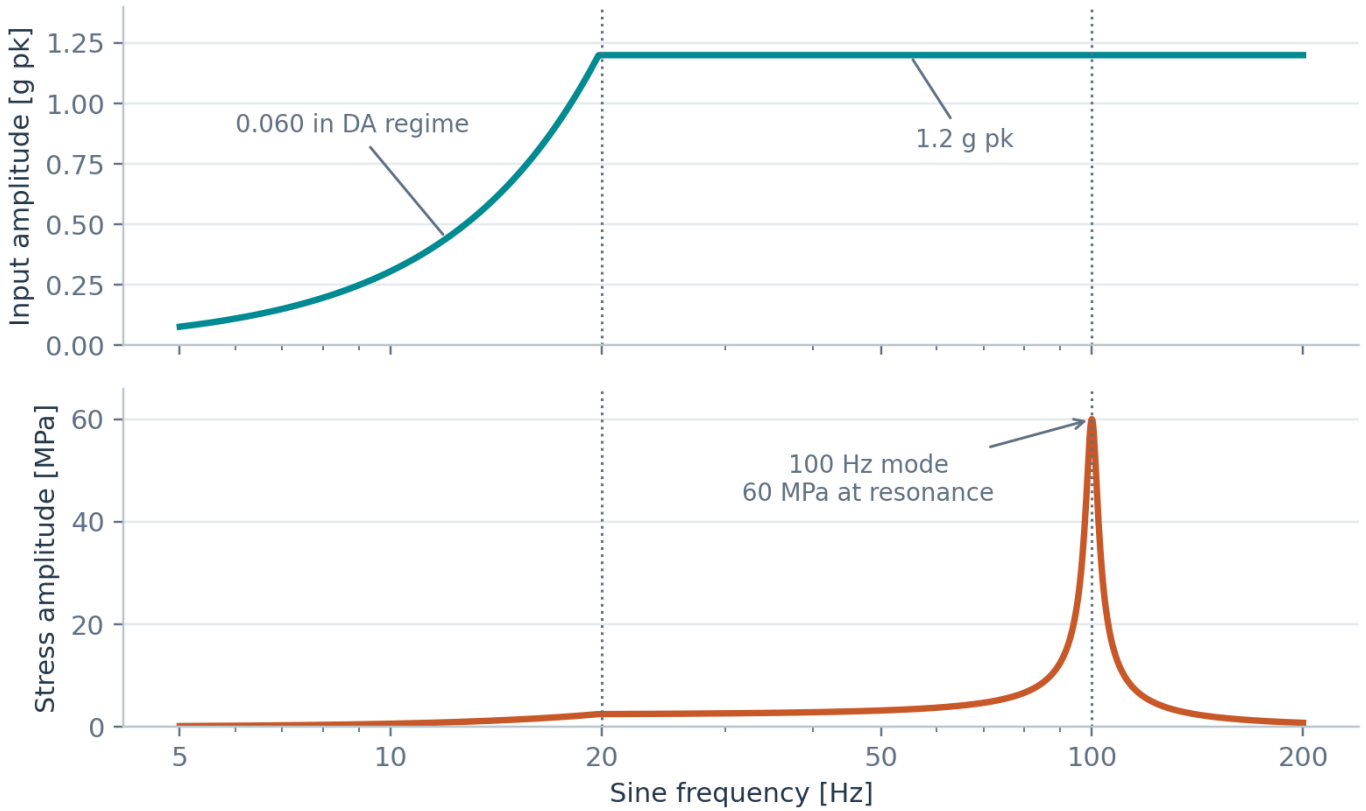


Figure 1. Original reconstruction of the illustrative input and steady-state stress response. The low-frequency displacement segment becomes an f² acceleration ramp. The stress response uses a single-mode FRF; the 60 MPa resonance is calculated, not measured.

A real reconstruction should preserve tolerances, sweep direction, number of axes and repeats, dwell logic, notching or limiting, controller definition, and whether the stated level is input, control average or response. If a dwell was triggered from a survey, use the actual dwell frequency and response history when available.

02 / PUT EVERY EXPOSURE ON CYCLES

One damage ledger, different counting rules

Adopt an amplitude-based S-N idealization $N(A)A^m = C$ and linear Palmgren-Miner accumulation. For a sine dwell with stress amplitude A_d, frequency f_d and duration T_d, the cycle count is deterministic:

$$D_{dwell} = (T_d\, f_d / C)\, A_d^{\,m}$$

For a quasi-steady sweep, integrate the changing response amplitude. If f changes logarithmically with $\beta = d(\ln f)/dt$, then $df/dt = \beta f$; the f in cycle rate cancels:

$$D_{sweep} = (1/C) \int A_\sigma(f)^m\, f\, dt$$
$$= (1/C\beta) \int A_\sigma(f)^m\, df$$

This elegant expression is conditional. It assumes the response follows the steady-state FRF. Check transient ring-up when damping is very low, sweep rate is fast, or the mode is low. Numerical integration of the equations of motion - with the actual frequency law, direction and controller history - is safer near that boundary.

For a stationary, Gaussian, narrow-band random stress response, Rayleigh peak statistics give

$$D_{random} = (T\nu_0/C)(\sqrt{2}\,\sigma_{rms})^m\, \Gamma(1 + m/2)$$

For broad-band or multimodal response, use an appropriate spectral method or synthesize enough realizations and rainflow-count the local stress. Random and sine S-N conventions must match. Stress range versus amplitude, mean stress correction, material/detail curve, and units must be explicit.

The constant C cancels only in a relative comparison using the same fatigue law and response quantity. It does not cancel differences in failure mode, mean stress, environment, joint state or material behavior.

03 / WORKED COMPARISON

The dwell owns this example

At the 100 Hz mode, 1.2 g peak produces a calculated 60 MPa alternating stress. With m = 5, the 60 s dwell contains 6,000 full cycles. Each up or down sweep lasts 99.7 s from 5 to 200 Hz, but it spends little of that time near the high-amplitude response.

Illustrative exposure	Local response	Damage / dwell
Sweep up, 2 oct/min	5-200 Hz, steady-state FRF	0.0193
Resonance dwell	60 MPa amp, 100 Hz, 60 s	1.0000
Sweep down, 2 oct/min	200-5 Hz, steady-state FRF	0.0193
Random comparison	25 MPa RMS, ν_0=100 Hz, 120 s	0.4722

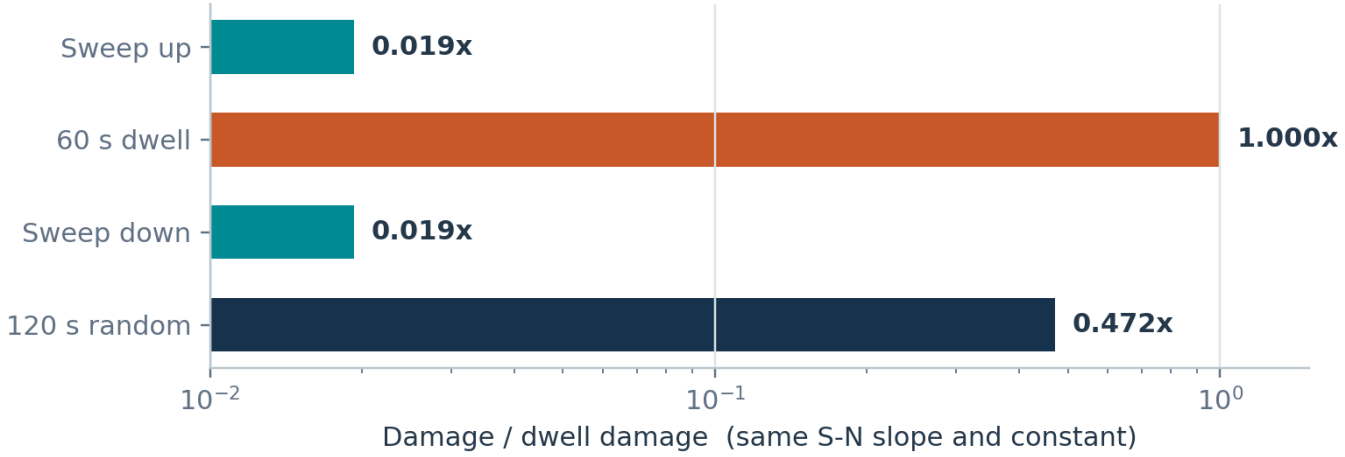


Figure 2. Original calculation normalized to the sine-dwell damage. Both sweeps together add 3.85% to the dwell in this linear, quasi-steady model. The random case is a separate 120 s exposure.

The complete sine sequence has 1.0385 times the dwell damage. A 120 s narrow-band Gaussian random response would need σ_{rms} = **29.27 MPa** to match it under the stated model. At 25 MPa RMS, the random case produces 45.5% of the complete sine-sequence damage.

That 29.27 MPa value is not an input conversion. It is an equivalent local stress response for this detail, mode, duration, statistical model and S-N slope. A second detail with another FRF can reverse the comparison.

04 / ENGINEERING APPLICATION

A defensible conversion workflow

1. Reconstruct the exposure. Digitize each segment: displacement or acceleration definition, peak convention, sweep law and rate, start/stop frequencies, directions, repeated runs, dwell frequencies and durations. Treat test-tolerance envelopes as bounds, not as the nominal history unless evidence says they were achieved continuously.

2. Choose the failure physics. Map both tests to the same response: local stress/strain, interface load, relative displacement or another mechanism-specific metric. An input accelerometer is not a fatigue surrogate unless the transfer to the critical detail is demonstrated.

3. Propagate uncertainty. Vary modal frequency, damping, joint state, boundary condition and stress recovery. A legacy dwell placed on a surveyed resonance can remain damaging after modest modal drift; a narrow random notch or band edge may not. Summarize the damage ratio across the credible set, not only at the nominal model.

4. Count by waveform. Use deterministic cycles for dwell, integrate or simulate sweeps, and use validated spectral fatigue or time-domain rainflow for random response. For measured nonstationary data, preserve event segmentation and time history where possible. Irvine discusses the added difficulty of random and nonstationary multiaxial counting [4].

5. Preserve non-fatigue objectives. A damage-equivalent random test does not reproduce sine peak response, resonance search, minimum-frequency evidence or nonlinear behavior. NASA’s sine-testing guidance emphasizes pre-test analysis, lower-level runs, response limits and aborts because the test deliberately passes through structural resonance [2].

6. Document the decision. State the S-N basis, amplitude convention, mean-stress treatment, Miner assumption, response model, random bandwidth model, test durations and uncertainty factors. Show both damage and peak-response comparisons. Record what remains unverified.

NASA-STD-7002B with Change 1 is an active payload-test standard revalidated March 24, 2023 [5]. Its scope is NASA payload test programs; it is not a universal conversion rule for launch vehicles, aircraft, marine hardware or a legacy contract. Program requirements and approval authority still govern.

05 / RETAIN AND APPLY

Three takeaways

1. Convert at the response. Sine amplitude, sweep rate and random PSD become comparable only after they are mapped to the same failure-driving quantity.

2. Separate sweep from dwell. A resonance dwell can dominate fatigue even when the sweeps occupy more clock time; compute each contribution and include repeats.

3. Damage equivalence is not test equivalence. Match fatigue without assuming that peak response, workmanship, nonlinearities or functional objectives are covered.

Reflection

If the random stress RMS in the example rises from 25 to 30 MPa while duration and statistics stay fixed, does it remain below the complete sine sequence?

Compact answer. No. Random damage scales with σ_{rms}^{m}. For m = 5, the factor is $(30/25)^5 = 2.488$. The new random damage is 1.13 times the complete sine-sequence damage. This sensitivity is why damping, response limits and stress-recovery uncertainty deserve more attention than a neat input-level conversion.

Sources and calculation notes

[1] Christensen, R. H., and Bellinfante, R. J. (June 1965). Some Considerations in the Fatigue Design of Launch and Spacecraft Structures. NASA CR-242. See pp. 31-49 for sine sweep, dwell and random-response treatment.

[2] NASA Engineering and Safety Center (September 2015). Technical Bulletin 15-03: Best Practices for Use of Sine Vibration Testing. Peer-reviewed NASA guidance on objectives, risks, sweep rate, response limits and aborts.

[3] Larsen, C. E., and Irvine, T. (VAL2015). A Review of Spectral Methods for Variable Amplitude Fatigue Prediction and New Results. Spectral moments, narrow-band damage and comparisons with rainflow.

[4] Irvine, T. (July 2016). Multiaxis Rainflow Fatigue Methods for Nonstationary Vibration. NASA NTRS record 20160010330.

[5] NASA (June 6, 2018; Change 1, March 24, 2023). NASA-STD-7002B w/Change 1: Payload Test Requirements. Official record and stated scope.

Verified October 1, 2026. Figures and table: original illustrative calculations. The 1965 report is used as a historical engineering source, not as a current governing requirement. Rounded display values may differ slightly from full-precision calculations.